How to read this student example

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The applied grading lens

Dimension	Question asked in grading
Decision	Does the memo directly answer the assigned policy question?
Economic mechanism	Does Krugman's economies-of-scale logic discipline the argument rather than appear as decoration?
Evidence	Do real, traceable data and exhibits establish the key claims, with assumptions and limits disclosed?
Professional readiness	Could this be sent to a Canadian policy audience tomorrow without substantial new analysis?

Each dimension was scored from 0 to 5 and weighted equally. The public example explains the rubric but does not disclose the student's individual grade.

Why this is a strong example

The memo gives a specific, conditional and reversible recommendation. It distinguishes international market access from local urban scale, turns sharing, matching and learning into observable policy conditions, and is unusually transparent about constructed scenarios, data limits and uncertainty.

Why it is not a perfect answer

A circulation-ready government memo would still need stronger causal evidence on welfare, validated decision thresholds, infrastructure and fiscal costing, and a clearer division of federal, provincial and municipal responsibilities. These are substantive next analyses, not cosmetic edits.

What to learn: answer first; make the economic mechanism do real work; separate facts from assumptions; state what the evidence cannot establish; and give measurable conditions under which the recommendation would pause, change or reverse.

Build housing and services before committing to the population target

To: Government of Canada | From: Anonymous ECON 342 student
ECON 342, Option B | October 21, 2026

Canada should consider reaching 100 million people by 2100 only if housing and services are ready before people move in and living standards per person improve. I recommend expanding several connected employment centres, called polycentric growth, while continuing to invest in the largest cities. Income gains must exceed any increase in shelter and commuting costs. The Government of Canada should fund this growth in stages through agreements that require housing and services to be delivered first. Delay the next stage of the population plan if these conditions fail.

Compare the proposal with continued growth under an official scenario

The alternative is continued population growth without the coordinated housing and service program proposed here. Statistics Canada's medium-growth scenario M1 supplies the population benchmark: about 41.7 million people in 2025 and 57.4 million in 2075. Across the published scenarios, the 2075 range is 44.0-75.8 million [1]. These projections depend on assumptions; they are not predictions or policy recommendations. M1 describes population, without prescribing housing investment. This memo therefore compares different population and investment plans together. It cannot identify the effect of population growth alone.

Reaching 100 million from 41.7 million over 2025-2100 requires (100/41.7)^(1/75) - 1, or about 1.17% annual growth. This constructed path reaches 74.7 million in 2075, when the official projections end. Only our path continues to 2100. If real GDP per person stayed unchanged, total real gross domestic product (GDP), meaning inflation-adjusted output, would be about 2.40 times its starting level. That would not show that each person was better off. The test is whether residents gain purchasing power after paying for shelter and travel to work.

Growth must create connected places with working services

Where people live determines those local costs. I compare two ways to allocate the same 58.3 million additional residents across Census metropolitan areas (CMAs). A CMA includes a core and nearby municipalities linked by commuting. The concentrated path sends 75% of additions to the six largest CMAs and 20% to eight selected mid-sized CMAs. The polycentric path sends 40% and 55%, respectively. Both send the remaining 5% to the rest of Canada and reach 100 million nationally. Each uses fixed 2021 CMA boundaries to prevent double counting. These allocations show where homes, jobs and services would be needed; they do not predict where people will move. The additions are net population growth, rather than immigrant admissions: births, deaths and net migration all affect the total [10].

Trade provides market access; nearby firms and workers provide urban gains

A larger market can spread fixed costs, which do not vary with output, across more sales or support more specialized products. Canadian exporters already reach foreign buyers, and imports already offer variety. I therefore expect the remaining gains to be strongest where customers, suppliers or infrastructure users need to be near one another. Market size, political boundaries and the concentration of people are different concepts.

Urban agglomeration means the productive advantages of locating near other firms and workers. Duranton and Puga explain three mechanisms: sharing, matching and learning [6]. Businesses can share specialized inputs and facilities. A larger pool of workers and vacancies can improve job matches. Contact can also spread knowledge. Housing and transport help workers reach these opportunities. Output per hour may rise, and competition and mobility may pass benefits to workers and consumers. The theory gives no numerical productivity estimate for Canada.

Additional centres need accessible jobs as well as homes

Established cities already have supplier networks and specialized jobs [6,7]. These advantages may weaken if housing becomes scarce or journeys become longer. Growing other centres could ease those pressures, provided they offer suitable jobs. The selected centres are Québec, Winnipeg, Hamilton, Kitchener-Cambridge-Waterloo, London, Halifax, Victoria and Saskatoon. Building within existing urban areas, with infrastructure and transport, should connect homes with jobs. Scattered subdivisions could instead duplicate facilities and lengthen trips. The density review supports weighing these benefits and costs; it does not show that smaller cities are always preferable [7].

The proposed agreements would therefore require infrastructure and staffed services before homes are occupied. Municipalities would zone serviced sites and approve homes. Provinces would coordinate water, wastewater, electricity, schools, health and regional transport. Federal funding would follow delivery milestones. Funding must cover construction, operation and maintenance, with enough skilled workers and equipment for delivery. Otherwise, projects could compete for crews while arrivals outpace capacity. Construction uses resources; financing determines how users and taxpayers pay. More construction GDP alone does not prove a lasting welfare gain.

The evaluation must measure benefits per person. Labour productivity is real output per hour; GDP per resident also depends on hours worked per resident. Neither measure fully captures housing costs, travel time or service access. Statistics Canada's February 2025 release reported that real GDP per person fell 1.4% in 2024 [4]. This historical result shows why total GDP is insufficient; it does not establish immigration's effect. Nor does a city-size correlation establish causation: workers and firms may choose places that are already productive.

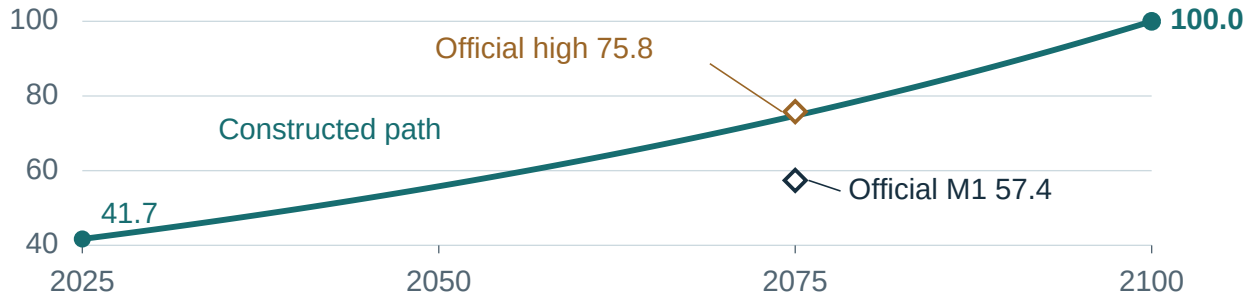
Age at arrival changes the timing of benefits and costs

Age at arrival also affects when benefits and costs occur. Younger working-age economic applicants may have more working years before retirement if suitable jobs exist. A Statistics Canada study linked younger age at landing to higher longer-term earnings among economic principal applicants, the main applicants in economic immigration programs. Canadian work experience and language also mattered [9]. This association does not establish a lifetime net benefit or an optimal age cutoff. An assessment should count children's schooling and future employment, older arrivals' skills and unpaid care, and health and pension costs. Housing and settlement needs should include accompanying family. Immigration cannot permanently prevent aging [10]. Economic admissions should therefore fit jobs and local capacity; family and humanitarian admissions also serve broader objectives.

The exhibits show how much capacity growth would require

Exhibit 1. A 100-million path is close to the 2075 high scenario

Canada | population, million persons



Source: Statistics Canada [1]; calculations from published rounded endpoints. Units: million persons, Canada. Official markers are 2075 endpoints, not annual trajectories. Only the constructed path extends to 2100.

Exhibit 1 shows the size of the demographic challenge. By 2075, the constructed path is near the official high scenario and well above M1. It indicates how many people would need housing and services, without showing an economic benefit. Exhibit 2 then keeps the national target fixed and changes where people would live.

Exhibit 2. Polycentric growth reallocates 20.4 million additions beyond the largest CMAs

Canada, million people	2025*	2100: C	2100: P
Six largest CMAs	19.76	63.48	43.08
Eight selected mid-sized CMAs	5.36	17.02	37.43
Rest of Canada	16.58	19.49	19.49
National total	**41.70**	**100.00**	**100.00**

C = concentrated; P = polycentric. Source: Census population [2,3] and chosen growth shares: 75/20/5% (C), 40/55/5% (P). *Synthetic 2025 base applies 2021 Census shares to 41.7 million; not observed 2025 CMA estimates. Within-group additions follow Census shares. Geography: 14 fixed 2021 CMAs plus the national residual. Rounded rows may differ from totals by 0.01 million.

Under the polycentric path, fewer additions go to Toronto, Montréal, Vancouver, Ottawa-Gatineau, Calgary and Edmonton, and more go to the eight selected centres. That mid-sized group becomes roughly seven times its synthetic 2025 size. This is a large expansion, so commitments should be staged rather than assumed easy or cheap. The research data show each CMA's allocation.

To translate these additions into homes, assume all added residents live in private households, with 2.4 people per additional household and 3% vacancy. This requires 24.3 million additional households and 25.0 million net dwellings. Annual need averages 334,000, rising from 210,000 in the first year of growth to 498,000 in the last. CMHC recorded 259,028 actual starts in 2025 [5a]; starts are not completed homes or net stock additions. Its separate 430,000-480,000 annual-start benchmark by 2035 aims to restore 2019 affordability under its own model [5b]. It does not describe this 2100 path. Our calculation excludes initial shortages, replacement and collective accommodation. These limits make the capacity conditions below essential.

The strongest objection is that the productive jobs may stay in the largest cities

The strongest objection is that building homes elsewhere may not bring productive jobs with them. Existing business networks cannot simply be recreated in another city. Building more homes within the largest cities could make better use of those networks. Subsidies to additional centres could instead leave buildings and infrastructure underused and reduce output per person. The scale of Exhibit 2 makes this risk serious. The polycentric preference must remain conditional: favour the largest cities where housing, services and access to jobs can expand at lower overall cost to the community [6,7].

Even if average living standards rise, some households could lose. In the 2021 Census, 33.2% of renter households and 14.8% of owner households spent at least 30% of income on shelter [8]. Temporary pandemic income supports affected these historical national figures; they are not current local baselines. Homes near jobs could benefit renters, younger households and newcomers. Existing owners could receive smaller house-price gains, while low-income residents could face displacement. Taxpayers and receiving communities could pay before benefits arrive. The agreements should therefore include public or nonprofit rental housing, relocation protection, transparent charges, negotiation with affected Indigenous governments, and screening for flood, wildfire, water and habitat risks.

Five measurable conditions determine whether growth should proceed

I propose five conditions. First, over the latest three years, each growing receiving CMA should add at least 1/0.97, or 1.031, net dwellings per net additional household. Address its initial housing shortage separately. Second, the share of renter households spending at least 30% of income on shelter should not rise from its pre-program baseline. Third, average one-way commute time should not exceed the local baseline. Fourth, 100% of new occupied homes should have certified water, wastewater, electricity and transport capacity, with funded school places and staffed health capacity meeting assessed demand before occupancy. Fifth, every five years, real GDP per person should exceed the published economic baseline for growth without this program. Inflation-adjusted disposable income after shelter costs should also not fall for the lowest-income fifth of local households. The housing ratio preserves the assumed vacancy allowance. The other thresholds protect access and require gains over the alternative. These are proposed rules, not official targets or estimated optimal thresholds.

Housing needs depend strongly on household size. Reducing assumed people per additional household from 2.4 to 2.0 raises average net dwelling needs from 334,000 to 401,000 a year. At a hypothetical average capacity of 350,000 net dwellings annually, enough homes could be added over the full period for the base case, but not for the smaller-household case. The latter would require delaying growth until capacity expands. Even the base case passes only a total-homes test: it does not show that homes arrive when and where needed. Initial shortages, replacement and new households formed by existing residents are excluded.

Canada should therefore expand the selected mid-sized centres in stages, subject to these five conditions. First publish how incomes and services are expected to develop without this program. Use M1's population benchmark through 2075 and state a separate assumption for later years. An independent review should compare results with that baseline and feasible growth in the largest cities, accounting for who settles where and changes affecting the whole economy. Pause expansion if housing or access conditions fail for two years in a row. Switch toward the largest cities if a five-year review finds that their higher income per person would outweigh the mid-sized centres' housing and service advantages. If neither approach improves living standards over the baseline, stop pursuing the 100-million target.

References and data notes

[1] Statistics Canada (2026). Population Projections for Canada, Provinces and Territories, 2025 to 2075, catalogue 17-20-0003, released January 27; printed p.6, first Canada bullet. Published 2025 base and 2075 LG/M1/HG totals, rounded to 0.1 million; underlying table 17-10-0057-01. No annual M1/HG interpolation is used.

[2] Statistics Canada (2024). Canada at a Glance, 2023: Population, tables 1-2, 2021 Census column. National count 36,991,981; cross-checks the six largest CMA populations. Census counts differ conceptually from postcensal estimates.

[3] Arbenser, L., Bernard, M.-C., Dormer, A., and Vipond, O. (2024). Short-term rentals in the Canadian housing market, Statistics Canada, catalogue 11-621-M, July 30, table C.1, printed p.16. Only the 2021 Census Population column is used; rental and dwelling columns are excluded. Fourteen selected CMAs plus Canada's residual define exhaustive, non-overlapping regions.

[4] Statistics Canada (2025). Gross domestic product, income and expenditure, fourth quarter 2024, February 28, printed p.1; per-capita series table 36-10-0706-01. Annual 2024 real GDP-per-person change, -1.4%, is a historical release vintage, not a current revised estimate. Volume measures use chained 2017 dollars; no nominal CMA GDP comparison is made.

[5a] CMHC (2026). Housing starts up 5.6% in 2025 from 2024, January 16, opening paragraph: 259,028 actual dwelling starts, Canada all areas. [5b] CMHC (2025). Updating Canada's housing supply shortages: new housing supply gap estimates, June 19. Annual starts benchmark by 2035, 430,000-480,000; model baseline 245,000-250,000. Different horizon and assumptions from this memo.

[6] Duranton, G., and Puga, D. (2004). Micro-foundations of urban agglomeration economies. Handbook of Regional and Urban Economics 4:2063-2117, sections 2-4. The linked author draft is dated June 11, 2003; the chapter was published in 2004. [7] Duranton, G., and Puga, D. (2020). The Economics of Urban Density. Journal of Economic Perspectives 34(3):3-26.

[8] Statistics Canada (2022), shelter-cost shares by tenure, September 21, table 4 and accompanying discussion, Canada row: owners 14.8%, renters 33.2% in 2021. Non-farm, non-reserve private dwellings; household income must be positive. These are household shares, not shares of people.

[9] Picot, G., Hou, F., Xu, L., and Bonikowska, A. (2022). Immigration selection factors and the earnings of economic principal applicants, Statistics Canada, June 22; sections on predictive factors and earnings differences. [10] Statistics Canada (2025). Canada's population estimates: Age and gender, July 1, 2025, September 24, pp.1, 3-4. Used for demographic accounting and aging mechanisms, not new numerical inputs.

Access date: October 21, 2026. During the table-download outage, official publications were accessed through preserved retrieval-service text captures, not complete CSV cubes; CMHC HTML files were downloaded directly. Raw JSON captures are unchanged retrieval responses, not original PDFs. The ZIP records exact source locations, transformations, hashes, assumptions and exclusions. Synthetic 2025 regional estimates hold 2021 shares constant; no claim of observed 2025 CMA populations is made. Household size, vacancy, allocation shares, capacity ceiling and policy thresholds are explicit assumptions, not official observations.

Reproducibility, AI use and individual contribution

Reproduction. The ZIP includes README.md, sources.md, raw and processed data, code, figures, the memo, verification_report.md, AI_LOG.md and CONTRIBUTIONS.md. Install Python and the two packages listed in the README, then run python code/build.py from the extracted folder. The code works offline from the saved source captures and CMHC HTML. It extracts the observations, checks the calculations, regenerates both exhibits and the six-page PDF, and exports the population allocations, growth path and housing sensitivity. The README explains how to repeat the analysis in a clean folder and inspect the rendered pages.

Audit scope. The checks cover source rows and columns, missing or duplicate Census observations, separate regional groups, official population comparisons, allocation totals, both 100-million endpoints, the housing sensitivity, unchanged raw-file hashes, page count and PDF links. The verification report records a clean-folder rebuild and visual inspection. The student confirms that they completed the required personal checks, including at least three source observations and one official aggregate. This confirmation is recorded separately from the automated checks.

AI disclosure. OpenAI Codex, based on GPT-6, was used through the desktop app with web retrieval, Python and ReportLab. AI assistance remains in the research design, source extraction, calculations, code, prose, exhibits, layout and audits. The student requested Option B, supplied the information required for the memo and confirmed permission to work alone. Important instructions were to separate facts from assumptions, check numerical denominators, preserve raw evidence and test the effect of household size. The AI log records retained work, errors, corrections and limits. AI output is not an economic source or independent certification.

Corrections. Some Statistics Canada downloads contained outage webpages instead of the requested data; these were excluded. Official publication text was saved as retrieval captures, with its narrower coverage disclosed. Annual official paths were not invented from endpoints. A CMA GDP comparison was dropped because matching data were inaccessible. The 2025 regional base is labelled synthetic. Starts are distinguished from completions and net stock, and future capacity is hypothetical. The audit distinguishes the 2003 author draft from its 2004 published chapter. Passing the housing-stock test does not establish a feasible delivery schedule.

Individual contribution. The student scoped the request, supplied the Option B brief, confirmed solo permission and requested revisions. No partner is claimed. Codex provided the documented research, drafting and coding assistance. The student confirms that they reviewed the retained argument, checked sources and arithmetic, reviewed the AI disclosure, and can explain the memo, code, figures and AI corrections. The student remains responsible for the final submission.

Submission. Pages 1-4 contain the memo, page 5 the references/data notes, and page 6 this appendix. The personal-check record is in STUDENT_FINAL_CHECKS.md. Canvas submission is a separate step: upload and reopen both final files. The brief gives Monday, October 26, 2026 at 12:00 p.m. Vancouver time; Canvas remains authoritative.